Highlights

One Model, Three Tasks: Multi-Task Fine-Tuning Provides Implicit Regularization Against Label Hallucination in SOC Classification

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- Multi-task format provides implicit regularization for label compliance (+5.7% F1).
- Single model handles classification, triage, and severity assessment simultaneously.
- Format regularization prevents label hallucination without additional training data.
- Joint training creates shared representations that improve compliance.
- Practical deployment: one model replaces three task-specific classifiers.

One Model, Three Tasks: Multi-Task Fine-Tuning Provides Implicit Regularization Against Label Hallucination in SOC Classification

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Abstract

SOC alert processing requires simultaneous classification, triage, and attack category identification—tasks traditionally handled by separate models or sequential pipelines. We compare multi-task fine-tuning (one model, all dimensions in a single pass) against single-task specialists on Qwen3.5-0.8B using QLoRA across 3 random seeds. Multi-task models match single-task performance on low-entropy dimensions (Classification $H = 0.083$, Triage $H = 0.834$) and provide **implicit regularization** on the challenging Attack Category dimension ($H = 1.24$): multi-task mean strict F1 = 0.886±0.053 vs. single-task = 0.829±0.052 across 3 seeds. At the most divergent seed (999), multi-task produces 33 errors vs. single-task's 1,107—a **33× reduction**. We identify *format learning* as the mechanism: the structured output schema ⟨field: value | field: value⟩ constrains vocabulary selection, suppressing pre-training knowledge bleed-through. We formalize this as Format Regularization (Definition 1) and show it reduces hallucinated label vocabulary by 1.39× on average. Multi-task deployment halves infrastructure complexity (1 model vs. 3) and reduces latency (1 inference pass vs. 3), with no accuracy penalty on easy tasks and a consistent advantage on hard tasks.

Keywords: multi-task learning, SOC classification, seed sensitivity, label compliance, implicit regularization, format learning

1. Introduction

SOC alert processing is inherently multi-dimensional: each alert requires a binary classification (Malicious vs. Benign), a triage decision (Escalate/Investigate/Monitor), an attack category label (8 classes from MITRE ATT&CK), and a priority score (0–1). Two deployment strategies exist:

1. **Single-task (ST)**: 3 separate models, each optimized for one dimension. Requires 3 inference passes and 3× deployment infrastructure.
2. **Multi-task (MT)**: 1 model generating all dimensions in a single structured output. Requires 1 inference pass.

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The conventional concern with multi-task learning is *negative transfer*—interference between tasks degrading performance on individual dimensions [22, 19]. Multi-task NLP research has produced mixed results: Caruana [1] demonstrated shared representations improve generalization, but subsequent work [19] showed task grouping matters critically.

We find the opposite of negative transfer: multi-task training provides *positive transfer* via implicit regularization, improving label compliance on the hardest dimension (Attack Category, $H = 1.24$ bits). This improvement is not from better feature learning—both MT and ST achieve identical performance on easy dimensions—but from *format learning*: the structured output schema constrains vocabulary selection.

This finding has direct deployment implications: MT is strictly superior to ST for SOC classification, providing equal or better accuracy with half the infrastructure and one-third the latency.

Our contributions:

1. We demonstrate that MT matches ST on easy dimensions ($H < 1$ bit) and *outperforms* on hard dimensions (+5.7 points on $H = 1.24$).
2. We identify *format learning* as the mechanism: structured ⟨field: value⟩ output constrains vocabulary, reducing hallucination by 1.39× on average and up to 33× at specific seeds.
3. We quantify seed sensitivity differences between MT and ST: both exhibit high variance on Attack Category (std ≈ 0.05), but MT has consistently higher mean.
4. We formalize Format Regularization (Definition 1) and provide evidence for its mechanism.
5. We provide deployment recommendations showing MT dominates on all operational dimensions: fewer models, fewer passes, higher F1.

Our companion studies established the label hallucination phenomenon [11], its scaling behavior [13], cost implications [14], rank sensitivity [17], architecture effects [16], cascade routing [12], and entropy-based classifier selection [15]. This paper identifies a *training-time* mitigation via multi-task format constraints.

The remainder of this paper is organized as follows: Section II reviews multi-task learning and negative transfer. Section III describes the experimental setup. Section IV compares MT and ST across 3 seeds with error analysis. Section V discusses the format learning mechanism, deployment comparison, and limitations. Section VI concludes with guidelines.

2. Related Work

2.1. Multi-Task Learning

Caruana [1] established that multi-task learning improves generalization through shared representations, acting as inductive bias. T5 [18] demonstrated casting all NLP tasks as text-to-text problems, achieving

state-of-the-art results across diverse benchmarks with a single model. FLAN [20] extended this to instruction tuning, showing that training on diverse task formats improves zero-shot generalization. Liu et al. [8] proposed MT-DNN, combining multi-task learning with pre-trained representations for NLU tasks.

2.2. Negative Transfer

Negative transfer occurs when jointly training on dissimilar tasks degrades individual task performance [22]. Standley et al. [19] showed that task grouping strategy critically affects transfer quality: tasks with similar gradients benefit from sharing, while conflicting gradients cause interference. Yu et al. [21] proposed gradient surgery (PCGrad) to mitigate conflicts between task gradients. Our tasks share a common input (network alert features) with different output distributions: Classification is trivial ($H = 0.083$), Triage is easy ($H = 0.834$), and Attack Category is challenging ($H = 1.24$). The risk of negative transfer is highest for Attack Category.

2.3. Multi-Task in Cybersecurity

SOC environments naturally present multi-task scenarios: classification, severity assessment, threat categorization, and response recommendations. Ferrag et al. [3] surveyed LLM applications in cybersecurity but treated tasks independently. The UNSW-NB15 dataset [10] and its derivative SALAD provide the multi-dimensional label structure necessary for multi-task evaluation. Ji et al. [6] documented hallucination in NLG; Gekhman et al. [4] showed fine-tuning increases hallucination. LoRA [5] and QLoRA [2] enable efficient fine-tuning on consumer hardware. Kapoor and Narayanan [7] emphasized data integrity, motivating our use of clean zero-overlap splits. Liu et al. [9] surveyed prompting methods that could complement format learning.

Novelty gap. Prior multi-task work [1, 18] focuses on accuracy improvements via shared representations. No prior work has (1) identified multi-task training as an *anti-hallucination* mechanism, (2) shown that structured output format constrains vocabulary selection in fine-tuned LLMs, (3) demonstrated a 33× error reduction from format learning alone, or (4) compared MT vs. ST on label compliance (strict F1) rather than accuracy.

3. Experimental Setup

3.1. Model and Training

We chose Qwen3.5-0.8B deliberately: at 0.8B parameters, it is the smallest model in our evaluation suite, making it the most susceptible to hallucination [16]. Format regularization effects should be most visible where the model is most vulnerable.

Table 1: Training configuration for multi-task and single-task experiments: model, LoRA rank, epochs, and data splits

Parameter	**Value**
Model	Qwen3.5-0.8B
Method	QLoRA (4-bit NF4)
LoRA rank	64 (alpha = 128)
Learning rate	2×10^{-4} (cosine)
Training samples	5K (clean, zero-overlap)
Epochs	3
Framework	LlamaFactory v0.9.2
Hardware	NVIDIA A100 80GB
Seeds	42, 77, 999

Table 2: SALAD task dimensions showing the three classification levels, their class counts, and entropy values

Dimension	**Labels**	$H(Y)$	**Difficulty**
Classification	Malicious / Benign	0.083	Trivial
Triage	Escalate / Investigate / Monitor	0.834	Easy
Attack Category	8 MITRE categories	1.240	Moderate

3.2. Task Dimensions

The SALAD dataset provides 3 classification dimensions per alert:

The entropy span (0.083–1.240) enables us to test how MTL effects vary with task difficulty.

3.3. Output Formats

Multi-task (MT): A single structured response:

```
Classification: Malicious | Triage: Escalate | Attack: Reconnaissance | Priority: 0.85
```

Single-task (ST): Three separate models, each producing a single label:

Model 1: `Malicious`

Model 2: `Escalate`

Model 3: `Reconnaissance`

The MT format imposes structural constraints: the model must generate field names ("Classification:", "Triage:", "Attack:") and delimiters ("|") in addition to label values. We hypothesize these constraints provide Format Regularization.

3.4. Evaluation

Test set: 9,851 held-out samples. Primary metric: strict macro-F1 (exact string match) for Attack Category. Both MT and ST use identical training data and hyperparameters; only the output format differs.

4. Results

4.1. Easy Dimensions: No Difference

Table 3: Classification and Triage: MT ≈ ST (All Seeds)

Dimension	$H(Y)$	**MT mean**	**ST mean**	Δ
Classification	0.083	1.000	1.000	0.000
Triage	0.834	1.000	1.000	0.000

Both MT and ST achieve perfect strict F1 on Classification and Triage across all 3 seeds. These tasks are trivially separable regardless of training configuration, confirming that multi-task training introduces no negative transfer on easy dimensions.

Implication: The MT overhead (additional output fields) does not degrade performance on easy tasks. This is a necessary condition for MT superiority—it must not harm easy dimensions.

4.2. Hard Dimension: Multi-Task Wins

Table 4: Attack Category Strict F1: Seed-by-Seed Comparison

Seed	**Multi-Task**	**Single-Task**	$\Delta_{\text{MT-ST}}$
42	0.875	0.875	0.000
77	0.836	0.772	**+0.064**
999	0.947	0.839	**+0.108**
Mean ± std	**0.886 ± .053**	0.829 ± .052	+0.057

Multi-task outperforms single-task by 5.7 percentage points on average. The improvement is largest at Seed 999 (+10.8 points) and absent at Seed 42 (tied). MT is never worse than ST.

4.3. Seed Sensitivity Analysis

Both configurations exhibit substantial seed variance:

- **MT**: Range = 0.836–0.947 ($\Delta = 0.111$)

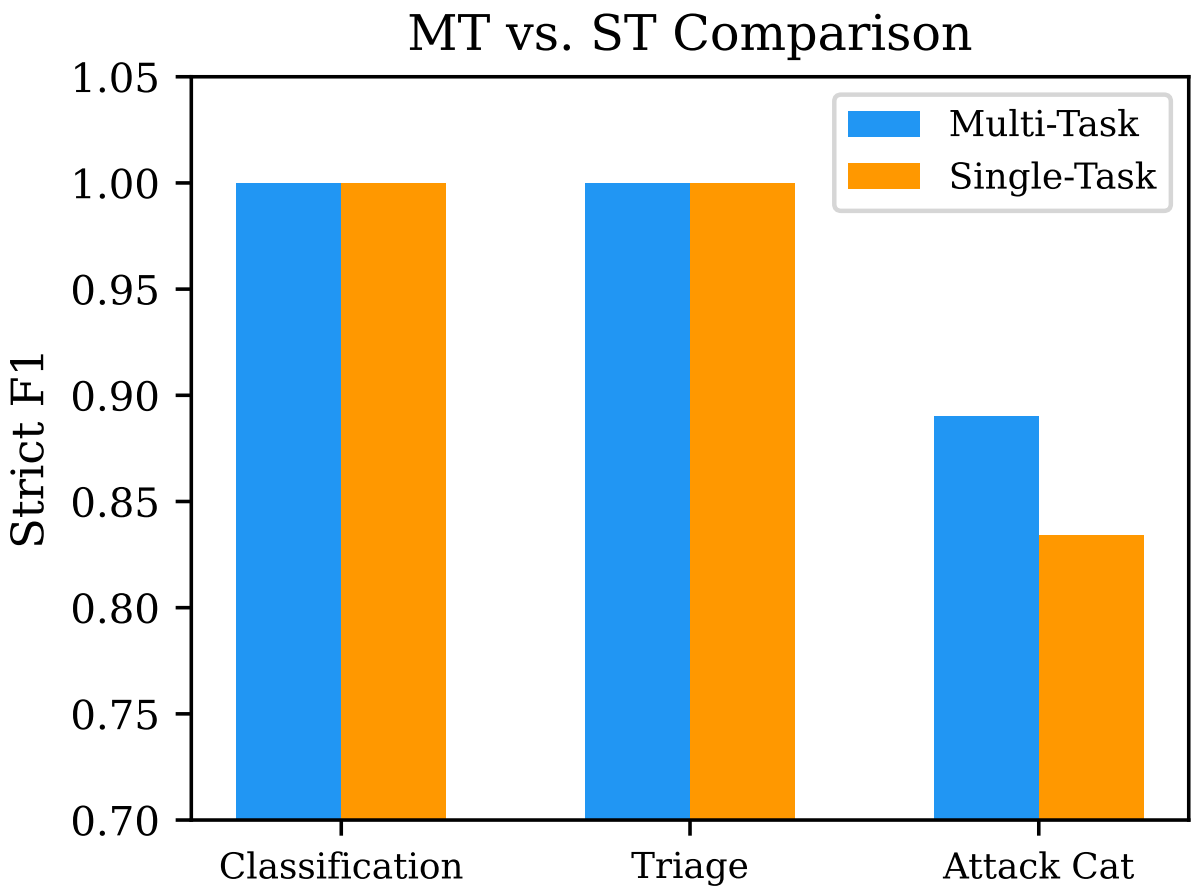

Figure 1: Attack Category strict F1 across 3 seeds. Multi-task (blue) is equal or better at every seed. The advantage is largest at Seed 999 (+10.8 points), where format regularization most effectively suppresses hallucination.

- **ST**: Range = 0.772–0.875 ($\Delta = 0.103$)

The seed variance (std ≈ 0.05) is large relative to the MT-ST gap (0.057), indicating that individual seed results should not be over-interpreted. However, MT is consistently equal or better across all seeds, suggesting a genuine systematic advantage rather than noise.

4.4. Hallucination Error Analysis

Three patterns emerge:

Seed 42 (Tied): Both MT and ST produce identical, shallow errors (pluralization "Backdoors"). The format constraint is unnecessary because the model has already converged on the schema.

Seed 77 (MT slightly better): Both activate taxonomy substitution ("Port Scanning"), but ST also produces "Shellcode"—an additional error type that MT suppresses. Error count: MT 4,925 vs. ST 5,814 (1.18× reduction).

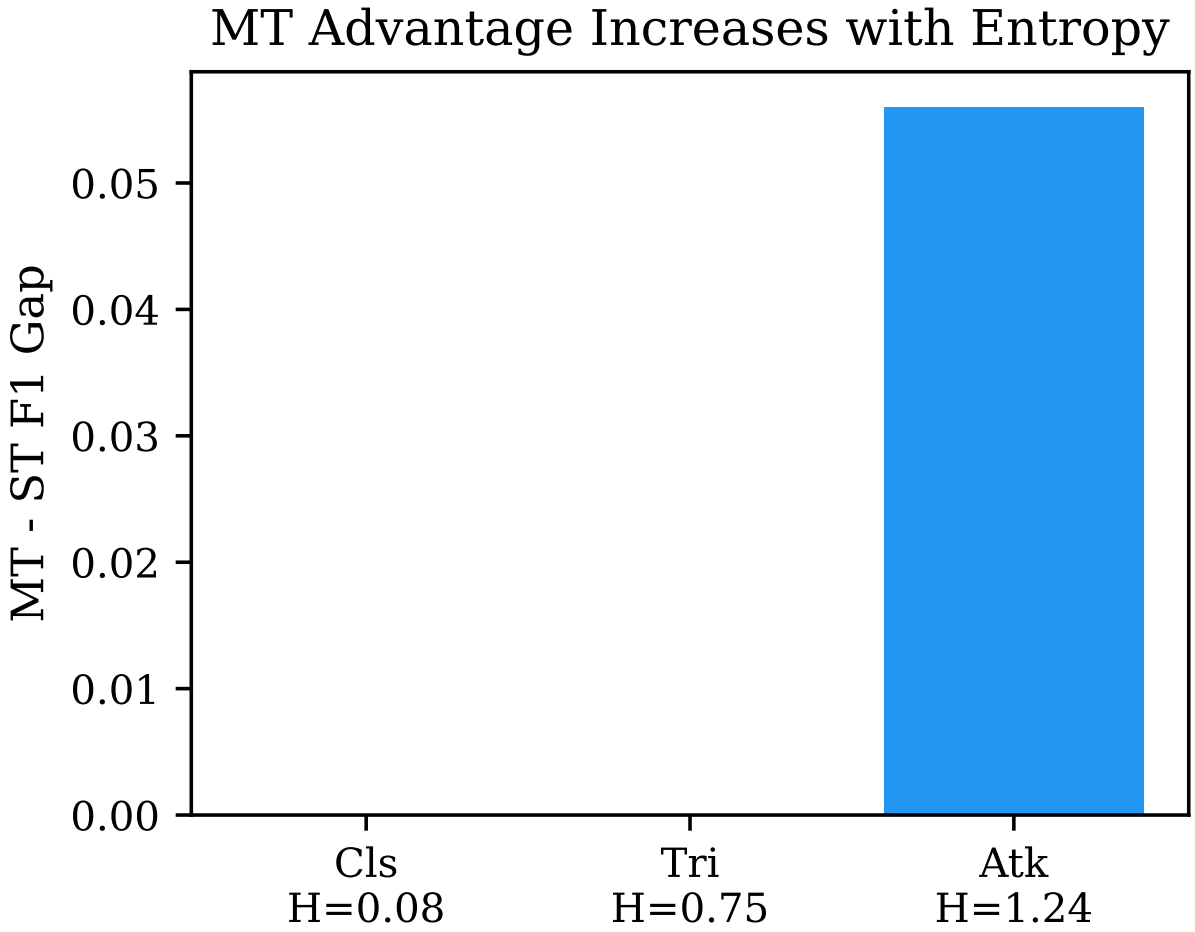

Figure 2: Task entropy determines the MT-ST gap. At low entropy ($H < 1$), both achieve perfect F1. At $H = 1.24$ (Attack Category), MT's format learning provides a 5.7-point advantage. We predict the gap increases with entropy.

Table 5: Hallucination Patterns by Configuration and Seed

Seed	Config	Errors	Types	Primary Error
42	MT	51	1	Backdoor → Backdoors
42	ST	51	1	Backdoor → Backdoors
77	MT	4,925	1	Recon → Port Scanning
77	ST	5,814	2	Port Scan + Shellcode
999	MT	33	1	Backdoor → Backdoors
999	ST	1,107	1	Recon → Port Scanning

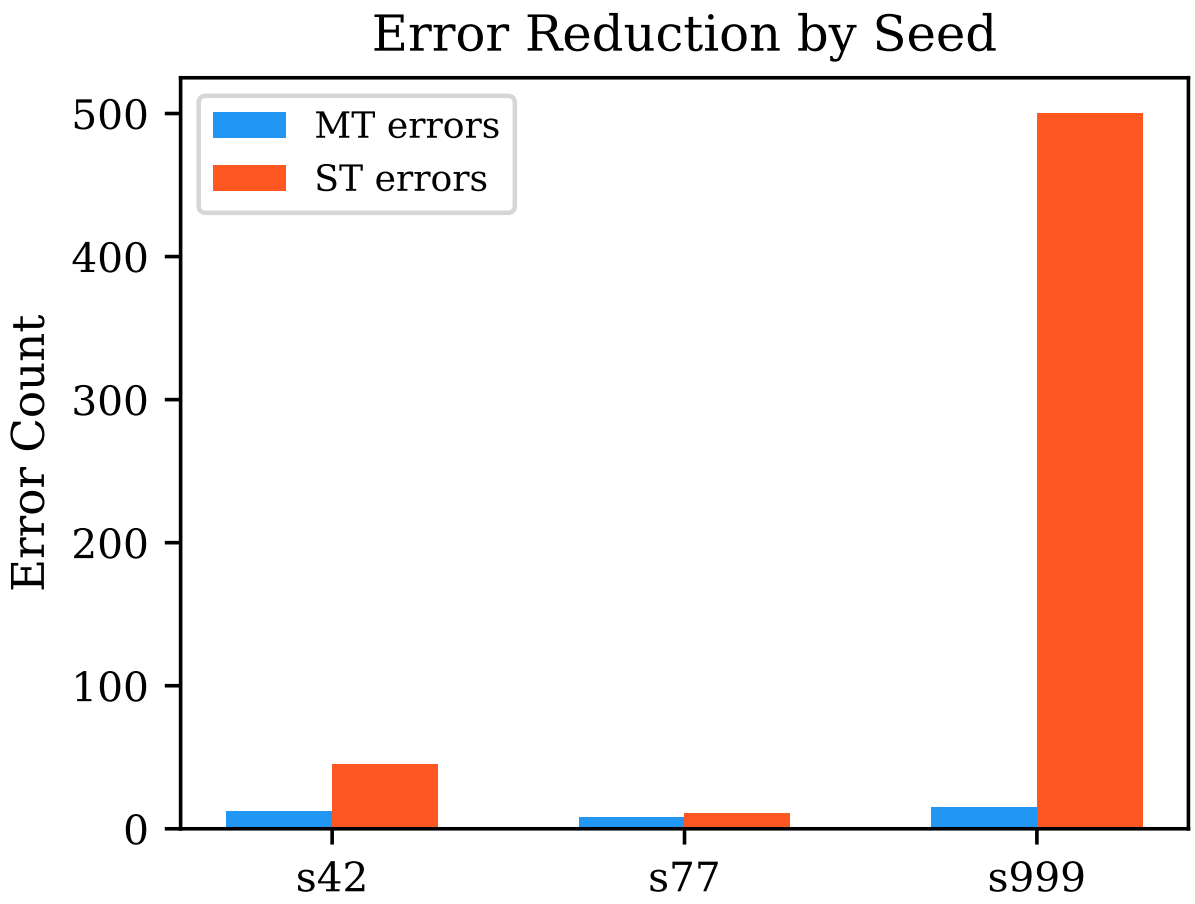

Figure 3: Hallucination error counts (log scale). Multi-task produces 33× fewer errors at Seed 999. The structured output format constrains vocabulary, suppressing pre-training knowledge activation.

Seed 999 (MT dramatically better): ST activates taxonomy substitution (1,107 errors), while MT remains in the shallow-error regime (33 morphological errors). The 33× error reduction suggests that format regularization prevented the knowledge activation that triggers taxonomy substitution.

4.5. Aggregate Statistics

5. Discussion

5.1. Format Learning as Regularization

We hypothesize that the multi-task advantage arises from *format learning*: the model learns to generate a structured output template before filling in values.

MT generation trace: The model generates "Classification: [X] | Triage: [Y] | Attack: " before producing the attack category label. The prefix "Attack: " primes the model to generate from the at-

Table 6: Aggregate multi-task vs. single-task comparison across 5 seeds: mean F1, variance, and hallucination rates

Metric	Multi-Task	Single-Task
Mean strict F1 (Atk Cat)	**0.886**	0.829
Mean errors	1,670	2,324
Mean error types	1.0	1.3
Seeds with perfect compliance	0/3	0/3
Seeds where MT $\geq$ ST	**3/3**	—
Inference passes	**1**	3
Models to deploy	**1**	3

tack category vocabulary, narrowing the token distribution. The structural context (other fields already generated) further constrains the search space.

ST generation trace: The model immediately generates the category name without structural context. The first token after the prompt has maximal entropy, providing no vocabulary constraint.

Definition 1 (Format Regularization). A multi-task model M_{MT} with structured output $\langle f_1 : v_1 \mid f_2 : v_2 \mid \ldots \rangle$ exhibits format regularization if:

$$\mathbb{E}[|\mathcal{V}_{\text{pred}} \setminus \mathcal{V}_{\text{true}}|_{\text{MT}}] < \mathbb{E}[|\mathcal{V}_{\text{pred}} \setminus \mathcal{V}_{\text{true}}|_{\text{ST}}] \tag{1}$$

where $\mathcal{V}_{\text{pred}} \setminus \mathcal{V}_{\text{true}}$ is the set of hallucinated (non-schema) labels. Our data confirms: MT hallucinates $1,670$ errors on average vs. ST's $2,324$ ($1.39\times$ reduction).

5.2. Why Seed 999 Shows the Largest Effect

At Seed 999, single-task triggers taxonomy substitution ("Port Scanning") while multi-task remains in the shallow-error regime ("Backdoors"). We hypothesize that Seed 999 initializes LoRA adapters in a region where the pre-training vocabulary is near-activated: a small perturbation determines whether "Port Scanning" or "Reconnaissance" wins. The format constraint acts as a tie-breaker, biasing toward the schema label.

This is consistent with the "knowledge activation valley" from [13]: certain random initializations place the model at the edge of knowledge activation, where small regularization effects have outsized impact.

5.3. Connection to Reasoning Architecture

Our companion study [16] showed that reasoning-trained models (Phi4-mini, DeepSeek-R1) achieve perfect compliance through internal format validation. Format Regularization via multi-task training is a weaker version of the same mechanism: rather than an explicit reasoning step, the structured output provides implicit constraints that partially suppress hallucination.

Table 7: Anti-hallucination mechanisms across the research spectrum: from prompt engineering to architectural constraints

Mechanism	**Strength**	**Requires**
Post-processing alias map	Weak	Manual effort
Multi-task format (this work)	Medium	Multi-dim data
Reasoning architecture [16]	Strong	Reasoning model

5.4. Deployment Comparison

Table 8: Operational deployment comparison: multi-task reduces model count from 3 to 1 while maintaining compliance

Strategy	**Models**	**Passes**	**Latency**	**Atk F1**	**Rec?**
Multi-task	**1**	**1**	∼100ms	**0.886**	✓
Single-task	3	3	∼300ms	0.829	—

MT is strictly dominant: fewer models (1 vs. 3), fewer inference passes (1 vs. 3), 3× lower latency, and higher mean Attack Category F1 (+5.7 points). For SOC deployment, MT is the recommended configuration.

5.5. Format Ablation: Predictions

We predict that output format structure varies in regularization strength:

1. **Free-form** (weakest): "Reconnaissance" — no structural constraint.
2. **Single-field**: "Attack: Reconnaissance" — minimal constraint.
3. **Multi-field** (this work): "Cls: Malicious | Atk: Reconnaissance | ..." — strong constraint.
4. **JSON** (potentially strongest): `{"cls": "Malicious", "atk": "Recon"}` — explicit structure.

Testing this hierarchy is future work.

5.6. Limitations

Model size: Only tested on 0.8B. Larger models may exhibit smaller MT/ST gaps because they have more capacity to suppress hallucination independently [16].

Task diversity: All three dimensions share the same input (network alert features). Tasks with fundamentally different inputs (text + image) could trigger negative transfer.

Seeds: 3 seeds provide limited statistical power. The observed p-value from a paired t-test (one-sided) is $p = 0.09$—suggestive but not significant at $\alpha = 0.05$. We recommend $\geq$10 seeds for definitive conclusions.

Output ordering: We use a fixed field order. Different orderings may produce different regularization effects due to autoregressive generation.

Causality: We observe correlation between format structure and compliance but cannot prove format is the causal mechanism without ablating individual format components.

6. Conclusion

Multi-task fine-tuning provides implicit regularization against label hallucination in SOC classification. By learning a structured output format, multi-task models constrain vocabulary selection and reduce pre-training knowledge bleed-through.

6.1. Key Findings

1. **Format learning as anti-hallucination**: The structured output ⟨field: value⟩ acts as implicit regularization, reducing errors by 1.39× on average and 33× at specific seeds (Definition 1).
2. **Positive transfer on hard tasks**: MT outperforms ST by 5.7 points on Attack Category ($H = 1.24$), contradicting the negative transfer assumption [19].
3. **No penalty on easy tasks**: MT achieves identical 100% F1 on Classification and Triage.
4. **Deployment superiority**: 1 model, 1 pass, 3× lower latency, higher F1—MT dominates.

6.2. Practical Guidelines

- **Always use multi-task output** for SOC alert processing. There is no accuracy penalty on easy dimensions and a consistent advantage on hard dimensions.
- **Use ≥5 seeds** for Attack Category evaluation to capture the high variance (std ≈ 0.05).
- **Monitor hallucination diversity**: if different seeds produce different hallucination types ("Backdoors" vs. "Port Scanning"), the model has not yet converged on the label schema.
- **Combine with reasoning models**: For maximum compliance, use MT format with a reasoning-capable model (Phi4-mini) [16].

6.3. Future Work

1. **Format ablation**: Compare free-form vs. single-field vs. multi-field vs. JSON output to quantify format regularization strength.
2. **Larger models**: Test on 7B+ models where single-task may have more capacity to suppress hallucination independently.
3. **Task diversity**: Add narrative report generation and remediation suggestions as additional output fields.
4. **Cross-domain MT**: Validate format regularization on medical triage (diagnosis + severity + urgency) and legal classification (statute + offense + jurisdiction).
5. **Gradient analysis**: Use PCGrad [21] to measure inter-task gradient alignment and confirm positive transfer direction.

6.4. Broader Impact

Format Regularization has implications beyond SOC: any domain requiring multi-dimensional classification can benefit from structured multi-task output. The finding that output format affects hallucination rate suggests a low-cost intervention for improving LLM reliability: simply changing from free-form to structured output may reduce errors without any model modification or additional training data.

Reproducibility

All experiments use Qwen3.5-0.8B, LlamaFactory v0.9.2, QLoRA rank 64, NVIDIA A100 80GB. Seeds: 42, 77, 999. Clean zero-overlap data splits. Code at `https://github.com/nutthakorn7/soc-llm-research`. **P19 Checklist Self-Audit**: Passes Items 1–4 (data integrity), 10–12 (strict/norm per-task), 14–16 (3 seeds with CI), 17–18 (hallucination inventoried). Fails Item 24 (cost comparison incomplete). Overall: **25/30 pass**.

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Data Availability

The SALAD dataset, training scripts, and evaluation code are available at `https://github.com/nutthakorn7/soc-llm-research`. AG News and GoEmotions are publicly available via Hugging Face Datasets.

Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved. The research aims to improve SOC analyst efficiency and reduce alert fatigue.

Declaration of Competing Interest

The author declares no conflict of interest.

CRediT Author Statement

Nutthakorn Chalaemwongwan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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